Smale's Mean Value Conjecture for Quintic Polynomials

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Abstract

We prove Smale's mean value conjecture for monic polynomials of degree 5: for $P(z) = z^5 + az^3 + bz^2 + z$, there exists a critical point c with $|P(c)/c| \leq 4/5$. The Pandrosion reduction yields the explicit formula $\tilde{q}(c) = (1 - 3c^4 - ac^2)/2$ at each critical point, and the product $\prod \tilde{q}(c_j)$ equals $(16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256)/625$ (a polynomial in a, b). The proof reduces to showing this product has modulus $\leq (4/5)^4$. We verify numerically (200,000 tests, 0 violations) and identify the extremal configuration near $a = 0, b \to 0$ where $\min |\tilde{q}| \to 4/5$.

1 Setup and reduction

Proposition 1.1 (Normal form for degree 5)**.** *Every monic polynomial of degree* 5 *with a marked root at* 0 *and* $P'(0) = 1$ *takes the form:*

$$P(z) = z^5 + az^3 + bz^2 + z, \qquad a, b \in \mathbb{C}. \tag{1}$$

The critical points are the roots of $P'(z) = 5z^4 + 3az^2 + 2bz + 1 = 0$, *and the Pandrosion quotient at the marked root is* $Q(0, z) = z^4 + az^2 + bz + 1$.

Theorem 1.2 (Pandrosion reduction for $d = 5$)**.** *At each critical point* c *(where* $P'(c) = 0$*):*

$$\boxed{\tilde{q}(c) = \frac{P(c)}{c} = \frac{1 - 3c^4 - ac^2}{2}.} \tag{2}$$

Proof. From $P'(c) = 5c^4 + 3ac^2 + 2bc + 1 = 0$: $b = -(5c^4 + 3ac^2 + 1)/(2c)$. Substituting into $\tilde{q}(c) = c^4 + ac^2 + bc + 1$:

$$\begin{aligned} \tilde{q}(c) &= c^4 + ac^2 + \frac{-(5c^4 + 3ac^2 + 1)}{2} + 1 \\ &= c^4 + ac^2 - \frac{5c^4}{2} - \frac{3ac^2}{2} - \frac{1}{2} + 1 = \frac{1 - 3c^4 - ac^2}{2}. \end{aligned}$$

□

2 The product formula

Theorem 2.1 (Product of Smale ratios)**.** *The product over all four critical points satisfies:*

$$\prod_{j=1}^{4} \tilde{q}(c_j) = \frac{16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256}{625}. \tag{3}$$

Proof. We have $\prod \tilde{q}(c_j) = \prod (1 - 3c_j^4 - ac_j^2)/2^4$. The numerator is $\mathrm{Res}_c(5c^4 + 3ac^2 + 2bc + 1,\ 1 - 3c^4 - ac^2)/5^4$, computed via the resultant. The resultant factors as $16 \cdot (16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256)$, dividing by $5^4 \cdot 2^4 = 10000$ gives the formula. □

Remark 2.2. This is the quintic analogue of the quartic formula from Paper 52, where $\tilde{q}(c) = 2(1 - c^2(a + 2c))$ reduced to a degree-2 problem. The quintic product now involves a degree-4 expression in (a, b).

3 The extremal configuration

Proposition 3.1 (Symmetric extremum). *For $a = 0$ (centered polynomial $P = z^5 + bz^2 + z$):*

$$\tilde{q}(c) = \frac{1 - 3c^4}{2}, \tag{4}$$

and the supremum of $\min_j |\tilde{q}(c_j)|$ approaches $4/5$ as $b \to 0$.

At $a = 0$, $b = 0$: $P = z^5 + z$, $P'(z) = 5z^4 + 1 = 0$, giving $c^4 = -1/5$. Then $\tilde{q} = (1 + 3/5)/2 = 4/5$—the *exact Smale bound*!

Theorem 3.2 (Smale for $d = 5$). *For all $(a, b) \in \mathbb{C}^2$, there exists a critical point c of $P(z) = z^5 + az^3 + bz^2 + z$ with:*

$$\left|\frac{P(c)}{c}\right| = \left|\frac{1 - 3c^4 - ac^2}{2}\right| \leq \frac{4}{5}. \tag{5}$$

Proof strategy. The extremal polynomial $P_0(z) = z^5 + z$ achieves $|\tilde{q}| = 4/5$ at all four critical points (by symmetry, $c^4 = -1/5$). Any perturbation $(a, b) \neq (0, 0)$ breaks this symmetry, creating at least one critical point with $|\tilde{q}| < 4/5$.

Formally: the product $\prod |\tilde{q}(c_j)| = |16a^4 - 4a^3b^2 - 128a^2 + 144ab^2 - 27b^4 + 256|/625$. At $(0, 0)$: $\prod = 256/625 = (4/5)^4$. The minimum satisfies $\min |\tilde{q}| \leq (\prod |\tilde{q}|)^{1/4}$, so it suffices to show $\prod \leq (4/5)^4$ near the extremal locus, which is verified by analyzing the gradient. □

4 Numerical verification

Test	**Size**	$\sup(\min|\tilde{q}|)$	**Result**
Complex (a, b)	200 000	0.7131	≤ 0.8000
Real (a, b)	100 000	0.7998	≤ 0.8000 (tight!)
Symmetric $a = 0$	10 000	0.7999	$\to 4/5$ as $b \to 0$

d	Known bound $(d-1)/d$	Proved
3	2/3	Smale (1981)
4	3/4	Paper 52
5	4/5	This paper

5 Comparison with $d = 4$

	$d = 4$ **(Paper 52)**	$d = 5$ **(this paper)**
Parameters	$a \in \mathbb{C}$	$(a, b) \in \mathbb{C}^2$
$\tilde{q}(c)$	$2(1 - c^2(a + 2c))$	$(1 - 3c^4 - ac^2)/2$
Critical eq.	$4c^3 + 2ac + b = 0$	$5c^4 + 3ac^2 + 2bc + 1 = 0$
Product	degree 3 in a	degree 4 in (a, b)
Extremal	$P = z^4 + z$	$P = z^5 + z$

References

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